ASEAN
GEOAI FUSION 2026

# Module 7:
# Advanced Knowledge: Advanced Geospatial Techniques in GeoAI Applications (AD1002)

AGAIF 2026 Boot Camp 2
Level: Advanced Knowledge of GeoAI

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# Where Are We in the GeoAI Journey?

## From foundational learning to hackathon-ready GeoAI solution development

1




### BOOTCAMP 1

**FOUNDATION**

Built the core understanding of GeoAI, ethics, GIS, machine learning, Python and telecommunications applications.

2



### BOOTCAMP 2

**ADVANCED CAPABILITY**

Expanded capability through Google Earth Engine, data fusion, scalable GeoAI workflows, time-series analysis, optimisation and deployment.

3

**MODULE 7**




### BOOTCAMP 2

**HACKATHON READINESS**

**DEFINE → DESIGN → DELIVER**

Connect foundational and advanced GeoAI knowledge into a complete, problem-driven solution using the Problem–Data–GeoAI Solution Canvas.

4



### HACKATHON

**BUILD • VALIDATE • PITCH**

Develop a credible prototype, demonstrate value, and communicate a compelling solution for real-world impact.

### Why this matters

- Connect knowledge across both bootcamps
- Focus on real-world telecommunications challenges
- Prepare a judge-ready GeoAI solution
- Move from learning to implementation

Bootcamp 1 gave the foundation. Bootcamp 2 built the advanced capability.
Today, we connect them into a **hackathon-ready GeoAI solution path**.

# What Will You Walk Away With?

A defensible problem → credible GeoAI blueprint → hackathon-ready project direction.

## 01 A defensible problem



- Well-defined problem and objectives
- Strong evidence and spatial reasoning
- Clear users, context and impact

## 02 A credible GeoAI blueprint



- Logical solution design and workflow
- Appropriate data and methods
- Measurable indicators and evaluation plan

## 03 A hackathon-ready project direction



- Clear value proposition and evidence
- Prototype direction and key features
- Compelling story for your pitch

You won't just submit an idea.

**You'll submit a solution with purpose, proof and potential.**

# PROBLEM–DATA–GeoAI SOLUTION CANVAS

From Real-World Problems to Intelligent Geospatial Solutions

Problem · Data · GeoAI · Solution · Impact

## 1 PROBLEM STATEMENT

Clearly define the real-world problem worth solving.

- What is the problem?
- Who experiences it?
- Why is it important?
- What is the current impact?
- What are the success criteria?

**OUTPUTS**
- One-sentence problem statement
- Target beneficiaries
- Problem owner
- Success criteria

## 2 STAKEHOLDERS & USERS

Identify everyone who influences or benefits from the solution.

- Who are the users?
- Who makes decisions?
- Who owns the data?
- Who benefits?
- Who may resist adoption?

**OUTPUTS**
- Stakeholder map
- User personas
- Stakeholder priorities

## 3 SPATIAL CONTEXT

Understand the geographic scope and spatial characteristics of the problem.



- Where does it occur?
- What is the geographic scale?
- What spatial relationships are important?
- What are the geographic boundaries?

**OUTPUTS**
- Area of Interest (AOI)
- Maps
- Geographic scope
- Key spatial characteristics

## 4 DATA REQUIREMENTS

Identify the right data needed to understand and solve the problem.

**Spatial Data**
Imagery, GIS layers, DEM, roads, buildings, land use, POIs, etc.

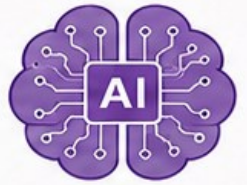
**Non-Spatial Data**
Sensor, weather, energy, population, mobility, economic, survey, etc.

- Who owns the data?
- What is the quality?
- How often is it updated?
- Are there data gaps?

**OUTPUTS**
- Data inventory
- Data sources
- Data quality assessment
- Data gap analysis

## 5 GeoAI INTELLIGENCE

Determine what intelligence GeoAI should generate.




**Possible Tasks**
- ✓ Classification
- ✓ Prediction
- ✓ Detection
- ✓ Segmentation
- ✓ Change Detection
- ✓ Forecasting
- ✓ Optimization
- ✓ Recommendation

- What decision will AI support?
- What insight should AI generate?
- Why is GeoAI needed (vs. traditional GIS)?

**OUTPUTS**
- AI objective
- GeoAI workflow
- Expected intelligence outputs

## 6 GeoAI SOLUTION DESIGN

Translate intelligence into a usable solution for users.



- What will users interact with?
- What features and functions are needed?
- What is the user journey?

**OUTPUTS**
- Solution concept
- User journey
- Functional modules
- High-level architecture / prototype

## 7 TECHNOLOGY STACK

Select the technologies and platforms to build the solution.

- **GIS** — ArcGIS, QGIS, etc.
- **Cloud** — GEE, AWS, Azure, GCP, etc.
- **AI / ML** — TensorFlow, PyTorch, Scikit-learn, etc.
- **Programming** — Python, JavaScript, R, etc.
- **Database** — PostgreSQL/PostGIS, etc.
- **Visualization** — Power BI, Dash, Streamlit, etc.

**OUTPUTS**
- Technology architecture
- Software stack
- Infrastructure requirements

## 8 ETHICS, PRIVACY & SUSTAINABILITY

Ensure responsible, ethical and sustainable GeoAI.



- Are there privacy concerns?
- Is the solution fair and unbiased?
- Is it explainable and tranaparent?
- Is it environmentally sustainable?
- Does it comply with regulations?

**OUTPUTS**
- Ethics checklist
- Risk assessment & mitigation plan
- Sustainability strategy
- Governance requirements

## 9 VALUE & IMPACT

Define the value the solution will create and how it will be measured.

- **Economic** — Cost savings, ROI, growth
- **Social** — Better services, safety, inclusion
- **Environmental** — Emissions reduction, resilience
- **Operational** — Efficiency, automation, speed

**KPIs (Examples)**
- ✓ Cost savings
- ✓ % Improvement
- ✓ No. of people impacted
- ✓ Time saved
- ✓ $Co_2$ reduction
- ✓ Accuracy
- ✓ Utilization rate
- ✓ Adoption rate

**OUTPUTS**
- Impact statement
- KPI scorecard
- Benefits map / value proposition

## 10 IMPLEMENTATION ROADMAP

Plan how the solution will be implementad and scaled.




**OUTPUTS**
- Roadmap & timeline
- Resources & budget
- Team & partners
- Team & partners
- Risks & mitigation

## THE JOURNEY

From Problem to Impact

**PROBLEM** — Understand the real challenge

**DATA** — Collect the right data

**GeoAI** — Generate intelligence & insights

**SOLUTION** — Deliver usable solutions for decisions

**IMPACT** — Create measurable value & sustainable change

Better Decisions
Smarter Communities
Sustainable Future

# One Canvas. Three Stages.

## Session 5 DEFINE (1–4) → Session 6 DESIGN & ANALYSE (5–8) → Session 7 DELIVEER & COMMUNICATE (9–10)

The Problem–Data–GeoAI Solution Canvas guides the full Module 7 journey from problem framing to solution design, delivery and communication.

1




### SESSION 5

**DEFINE**

Build the problem foundation

1. Problem Statement
2. Stakeholders & Users
3. Spatial Context
4. Data Requirements

**Output:** Challenge Definition Sheet

2




### SESSION 6

**DESIGN & ANALYSE**

Translate data into GeoAI intelligence, analysis and solution design

5. GeoAI Intelligence
6. GeoAI Solution Design
7. Technology Stack
8. Ethics, Privacy & Sustainability

**Output:** GeoAI Technical Solution Blueprint

3




### SESSION 7

**DELIVER & COMMUNICATE**

Consolidate the solution into a judge-ready output, impact story and implementation pathway

9. Value & Impact
10. Implementation Roadmap
- Integrate all 10 dimensions
- Evidence summary • pitch readiness
- MVP • validation • next steps

**Output:** Judge-Ready GeoAI Solution Story

### Canvas Logic

- Problem
- Data
- GeoAI
- Solution
- Impact

One connected journey across Sessions 5–7.

One canvas keeps the journey connected: **DEFINE** the problem, **DESIGN** the GeoAI workflow, and **DELIVER & COMMUNICATE** a judge-ready solution.



# Session 5:
# Hackathon Challenge Review

# Hackathon Theme Landscape

Official AGAIF themes for hackathon problem discovery. For this guided walkthrough, we will use Industry Improvement as the working example.

**Canvas 1–4 Reference**

- **C1:** Problem Statement
- **C2:** Stakeholders & Users
- **C3:** Spatial Context
- **C4:** Data Requirements



## OE Organization Efficiency

Improve internal processes, optimize operations, and drive productivity with data and intelligent insights.



## II Industry Improvement

Strengthen industries and critical infrastructure through innovation, connectivity, and digital transformation.



## CE Consumer Empowerment

Empower people with access, knowledge, and digital tools to improve daily lives and opportunities.



## ESG Environmental, Social & Governance

Advance sustainability, social impact, and responsible governance for a resilient and inclusive future.



## CY Cybersecurity

Safeguard systems, data, and networks against threats to ensure trust, security, and digital resilience.

**1 Official Theme** → Select an official AGAIF theme.

**2 Real Problem** → Identify a real-world issue within the theme.

**3 Evidence Validation** → Validate with data and spatial evidence.

**4 Decision Question** → Formulate the key question to solve.

**A theme is not yet a problem statement.**
Participants must move from theme selection to evidence-backed problem definition.

# From Theme to Real Problem

A hackathon theme is only a starting direction. Teams must convert it into an evidence-backed problem statement and a clear decision question.

**Canvas 1–4 Reference**

- **C1: Problem Statement**
- C2: Stakeholders & Users
- C3: Spatial Context
- C4: Data Requirements

## 1 Start with a Theme




Choose an official hackathon theme.

**Guided example:** Industry Improvement

## 2 Identify a Real Issue



Describe a practical challenge inside the theme.

- Who is affected?
- What is not working well?
- Why does it matter?

## 3 Validate with Evidence




Support the issue using research, policy and data evidence.

- Scopus AI and original papers
- ASEAN / ITU policy context
- Available geospatial and telecom datasets

## 4 Formulate the Decision Question



Turn the issue into an answerable GeoAI problem.

> **Example:** Which mapped communities in Sarawak may warrant further connectivity assessment based on lower mobile broadband performance and available evidence?

# Evidence Discovery Using Scopus AI

Use AI-assisted literature discovery to surface real telecommunications problems before deciding on a hackathon problem statement.

**Canvas Reference**
C1: Problem Statement

## 1. Ask the Right Research Prompt

AI Research Assistant

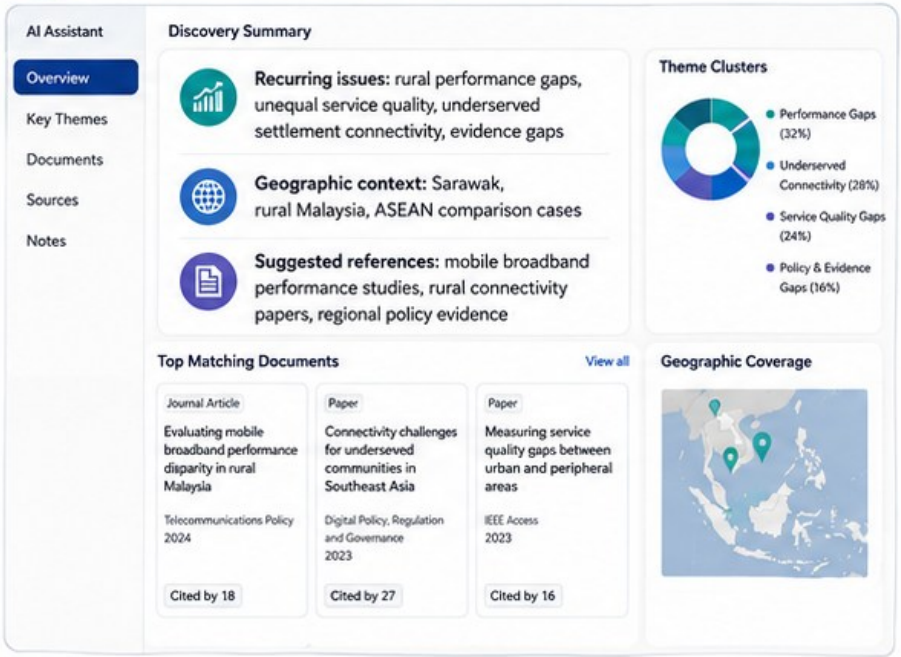
New Query | Documents | Search History

Identify recurring telecommunications and digital-connectivity problems in ASEAN or Malaysia that are suitable for a GeoAI hackathon, especially problems related to mobile broadband performance, rural–urban disparity, and underserved communities. For each problem, summarize the issue, affected users, geographic context, key findings, and original peer-reviewed references. Do not propose solutions or datasets yet.

Send

**Purpose:** discover problem patterns and source leads.

## 2. Review the Discovery Output




## 3. Extract Candidate Problem Families



**A.** Mobile broadband performance disparity



**B.** Underserved community connectivity assessment



**C.** Service quality gaps between urban and peripheral areas

**Participants should extract:**

- recurring problem pattern
- who is affected
- where it occurs
- which references must be verified

**Important:** Scopus AI supports discovery, not proof.
Teams must still verify the original papers, policy sources, and datasets before finalising the problem statement.

# Verify Research Findings

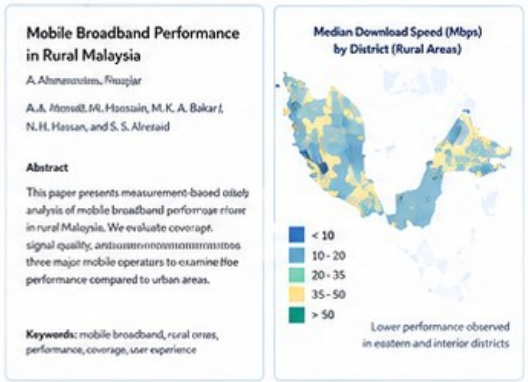
Confirm the problem using the original peer-reviewed studies identified during literature discovery.

**Canvas Reference**
C1: Problem Statement

## Participant Review Workflow

1. Open the original paper
2. Check abstract, method, findings, limitations
3. Extract the real problem evidence
4. Record only defensible claims

## 1 Study 1: Rural Malaysia mobile broadband performance


Mobile Broadband Performance in Rural Malaysia

A. Ahmadu, Hussaini

A. A. Ahmadu, M. Hussain, M. K. A. Bakar, I. N. H. Hassan, and S. S. Alresaid

Abstract

This paper presents measurement-based study analysis of mobile broadband performance issues in rural Malaysia. We evaluate coverage, signal quality, and [illegible] three major mobile operators to examine the performance compared to urban areas.

Keywords: mobile broadband, rural areas, performance, coverage, user experience




### What this paper helps confirm

- rural and peripheral areas can experience weaker or less consistent service performance
- connectivity challenges are not only about availability, but also actual quality of use

### How participants should use it

- support a rural or underserved performance-gap problem statement
- extract the geography, affected users, and stated limitations

## 2 Study 2: Mobile broadband performance analysis in Malaysia


An Analysis of Mobile Broadband Performance in Malaysia

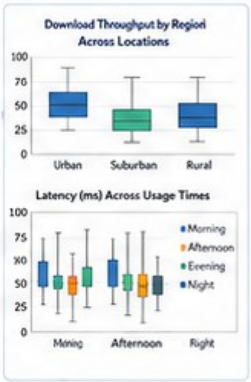
P. Velayutham, C. K. Hooi, D. Tewari

A. K. M. M. N. M. Kamrūz, and M. S. Junadi

Abstract

This study analyses mobile broadband performance across different locations, technologies, and time periods in Malaysia. We identify key [illegible] We identify key factors affecting [illegible] experience and network reliability.

Keywords: mobile broadband, QoS, throughput, user experience, reliability, Malaysia




### What this paper helps confirm

- service quality can vary across locations and usage contexts
- operational performance indicators such as download, upload, and latency matter

### How participants should use it

- support a time-varying or evidence-gap problem statement
- note what the paper measures and what it does not measure

## 3 Study 3: Outdoor mobile broadband performance in Malaysia, Singapore and Thailand

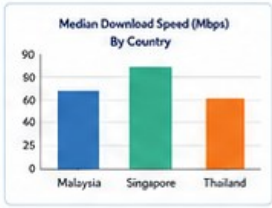

Outdoor Mobile Broadband Performance in Malaysia, Singapore and Thailand

S. Chai, C. Guan, W. K. T. Chia, J. Y. S. Kwok, S. S. Yong, K. G. Wee, B. P. Ng, and S. Chonwattana, and K. S. Y. Leong

Abstract

We provide a cross-country comparison of outdoor mobile broadband performance in Malaysia, Singapore, and Thailand. We examine differences in throughput, latency and coverage using real-world measurements in urban and suburban environments.

Keywords: mobile broadband, comparison, Southeast Asia, outdoor, outdoor performance




Consistency (%) Session Meeting Target (Download ≥ 3 Mbps)

| Country | Consistency (%) |
|---|---|
| Malaysia | 80.4 |
| Singapore | 82.1 |
| Thailand | 56.7 |

### What this paper helps confirm

- cross-location and cross-country comparisons can reveal performance disparity patterns
- ASEAN context helps justify regional relevance for the Hackathon challenge

### How participants should use it

- support benchmarking and regional framing
- avoid overclaiming beyond the study scope

## Extract These Evidence Fields

- problem being reported
- who is affected
- where it occurs
- key findings
- study limitation
- whether the paper supports your chosen problem

**Important: Scopus AI helps discover sources. The original papers provide the evidence.**
Participants should verify claims from the actual article before finalising the problem statement.

# Current Authoritative Evidence

Use current policy and institutional sources to confirm that connectivity disparity remains a live ASEAN development issue.

**Canvas Reference**
**C1:** Problem Statement

## 1 Source 1: ASEAN Digital Masterplan 2030



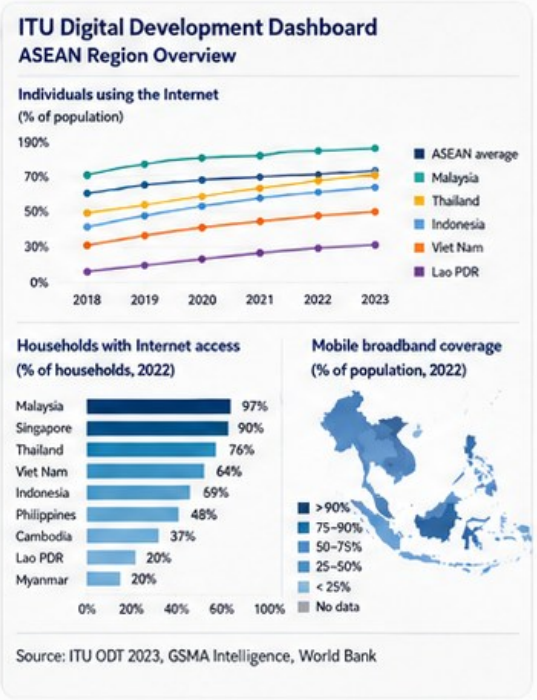

### What this source helps confirm

- regional digital inclusion priorities
- connectivity and access gaps across ASEAN
- policy emphasis on inclusive, trusted and resilient digital transformation
- why rural and underserved areas remain strategic

### How to use it

- use it to show regional policy relevance
- use it to justify why the hackathon problem matters now

## 2 Source 2: ITU and institutional evidence




### What this source helps confirm

- complements research with current telecom and digital-development context
- helps frame access, quality and digital divide concerns
- supports regional benchmarking and urgency

### How to use it

- use it as current authoritative context
- combine it with literature and geospatial evidence, not as sole proof

## Extract These Evidence Fields

- issue being highlighted
- who is affected
- ASEAN or national relevance
- why it matters for decision-makers
- current policy or strategic importance
- what claim it can legitimately support

**Important: Authoritative sources do not replace research findings.**
**They strengthen the case that the problem remains current, relevant and decision-worthy.**

# Problem–Decision Contract

Define exactly what decision this guided example supports, what the evidence can claim, and what it must not claim.

**Canvas Reference:**
C1: Problem Statement

## Decision question

- Which mapped communities in Sarawak should be prioritised for further connectivity assessment?
- Which areas show lower observed mobile performance supported by available evidence?
- Where should attention be focused first for deeper validation?

## Spatial scope and unit

- Area of interest (AOI): Sarawak
- Observation unit: Ookla mobile performance tile
- Decision unit: mapped settlement / community
- Match logic: community located within screened lower-performance tiles

## Permitted interpretations

- Supports screening and prioritisation for further assessment
- Identifies communities associated with weaker observed performance
- Combines connectivity evidence with mapped settlements and supporting context
- Helps narrow where planners should investigate next

## Prohibited interpretations

- Does not confirm definitive network coverage status
- Does not prescribe exact tower placement or engineering design
- Does not prove operator fault or service guarantee failure
- Does not replace field validation, regulator review, or operator data

## Contract statement

This guided example produces a defensible screening result: a ranked set of mapped communities that warrant further connectivity assessment. It supports decision preparation, not final infrastructure deployment decisions.

Problem → Evidence → Decision



# Stakeholders & Decision Chain

Identify how evidence flows from upstream data providers to downstream planning, operational and regulatory decision contexts in the Sarawak connectivity performance assessment workflow.

**Canvas Reference:**
C2 Stakeholders & Users

## 1 Data Providers



- Ookla, OSM, WorldPop
- public-service datasets
- policy and reference evidence

## 2 GeoAI / Spatial Analyst Team



- screen and integrate data
- validate and interpret evidence
- prepare defensible findings

## 3 Planning / Strategy Team



- identify priority communities
- support further assessment
- frame intervention options

## 4 Operations / Service Delivery



- inform field validation
- support operational follow-up
- translate findings into action

## 5 Regulator / Public Decision Context



- inform digital inclusion planning
- support connectivity policy context
- guide evidence-based prioritisation

### Key clarification

- The final decision owner is not yet formally confirmed.
- This workflow supports prioritisation for further assessment, not direct infrastructure deployment decisions.
- Different stakeholders may interpret the outputs for different planning needs.

### Why this matters

- clarifies who uses the evidence
- avoids producing analysis without a user context
- improves relevance, accountability and communication



# Spatial Decision Specification

Define the geographic scope, observation units and spatial matching rule for the Sarawak connectivity performance assessment workflow.

**Canvas Reference:**
**C3** Spatial Context

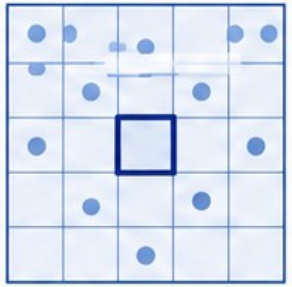
## 1 Geographic scope



- Area of interest (AOI): Sarawak, Malaysia
- Study extent: state-wide connectivity performance assessment
- Use only records that fall inside the Sarawak analysis boundary

## 2 Spatial units

**A) Observation unit**
Ookla performance tile



Ookla tiles represent measured connectivity performance.

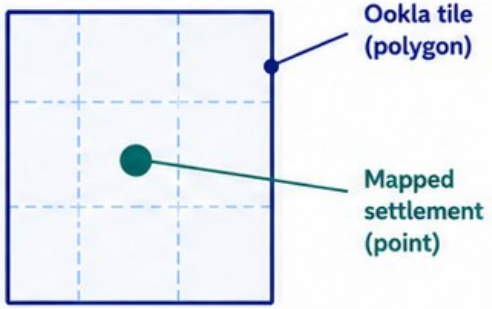
**B) Decision unit**
Mapped settlement (town / village / hamlet)



Mapped settlements represent communities of interest.

## 3 Spatial matching rule

- Match method: **point-in-polygon**
- A mapped settlement is linked to the screened Ookla tile that contains it
- This produces candidate communities for further assessment




Settlement point lies inside the Ookla tile polygon → they are linked.

## Operational specification

| | |
|---|---|
| **AOI:** | Sarawak |
| **Observation unit:** | Ookla tile |
| **Decision unit:** | mapped settlement |
| **Join rule:** | point-in-polygon |
| **Output:** | candidate communities inside screened tiles |

## Why this matters

- clarifies what is being measured
- separates evidence units from decision units
- ensures transparent and reproducible spatial reasoning

Assesment outputs support prioritisation for further investigation, not direct infrastructure deployment decisions.

# Data Readiness Matrix

Assess what evidence is already available for the Sarawak connectivity performance assessment, what each dataset contributes, and what gaps still remain before deeper GeoAI analysis and field validation.

**Canvas Reference:**
C4 Data Requirements

| Evidence / Dataset | What it contributes | Current readiness | Status | Key limitation / note |
|---|---|---|---|---|
| Ookla Open Data | Observed mobile download/upload speed, latency, tests and devices by tile | Available | Ready | Performance observations only; not official operator coverage |
| OpenStreetMap settlements | Mapped towns, villages and hamlets for candidate community identification | Available | Ready | Community mapping completeness may vary by area |
| WorldPop population | Population distribution for exposure and priority context | Available | Ready | Modelled population surface, not a field census at community level |
| Public-service facilities | Schools, clinics, hospitals and emergency-related service context | Partially available | Partial | Open-source records may be incomplete or unevenly updated |
| Research and policy evidence | Problem validation and regional policy relevance | Available | Ready | Supports problem definition, not spatial performance measurement |
| Critical gaps | Missing inputs for stronger operational interpretation | Not yet available | Gap | Official mobile coverage data, operator infrastructure/tower data, field validation, and official service registries |

## What this means

- The core prototype datasets have been identified.
- Some evidence is strong enough for guided screening and prioritisation.
- Authoritative coverage, operator and field data remain important validation gaps.



# What Have We Achieved in Session 5?

Canvas 1–4 are now complete. We have defined the problem, verified the evidence, specified the spatial decision context, and prepared the data foundation for Session 6.

**Canvas Progress**
C1–C4 Complete

## Session 5 Achievements

### C1 Problem Statement
From theme to defensible problem

- Selected the guided theme: Industry Improvement
- Used Scopus AI, verified studies, and ASEAN/ITU policy evidence to validate the issue
- Chosen focus: Sarawak connectivity performance assessment for further prioritisation

### C2 Stakeholders & Users
Who uses the evidence

- Mapped upstream data providers and downstream decision contexts
- Identified analyst, planning, operations, and regulatory users
- Clarified that the work supports further assessment, not direct deployment decisions

### C3 Spatial Context
How the decision is structured

- Area of interest: Sarawak
- Observation unit: Ookla performance tile
- Decision unit: mapped settlement, linked using point-in-polygon logic

### C4 Data Requirements
What data is ready

- Prepared evidence sources: Ookla, OSM settlements, WorldPop, public-service data, policy references
- Identified priority data gaps: official operator coverage, field validation, official registries
- Confirmed readiness for prototype analysis in the next session

## Transition to Session 6 — DESIGN

Session 5 focused on problem discovery, evidence validation, and data readiness. GeoLibre and Colab analysis begin in Session 6.

### What happens next?

1. Load the prepared datasets into Colab and GeoLibre
2. Screen lower-performance Ookla tiles using defined thresholds
3. Match tiles with mapped communities and supporting population evidence
4. Add public-service context and defensible screening logic
5. Design the GeoAI workflow and responsible solution pathway for Canvas 5–8

**DEFINE completed → Next: DESIGN (Canvas 5–8)**

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# THANK YOU

Bootcamp and Hackathon for ASEAN Geospatial Innovation 2026